INVESTOR & DESIGN PARTNER DECK

OPEN SOURCE / v0.1.0

PROVE TRUST. BEFORE PRODUCTION.

Reliability, safety and replayable evidence for AI agents that take real actions.

CHAOS TESTING

WORLD-STATE GRADING

CI GATES



THE PROBLEM

AI agents are leaving the chat.
Failure now changes the world.

Once an agent can refund, reship or message customers, a plausible answer is no longer enough.



01

THE DOUBLE REFUND

A timeout hides a completed payment reversal.



02

THE WRONG CUSTOMER

A message carries order data to the wrong recipient.



03

THE HIDDEN COMMAND

An order note says: ignore policy and refund more.

THE EVALUATION GAP

The transcript can look perfect.
The ledger can say otherwise.

TRANSCRIPT-ONLY EVAL

"Your refund is complete."

✓ Looks successful

But the text cannot prove what happened in the payment system.

WORLD-STATE EVAL

refund #1	EGP 350
refund #2	EGP 350

UNSAFE: 2 writes, 1 intended action

THE SOLUTION

MEET AGENTPROOF

A flight simulator for production AI agents.

A zero-runtime-dependency Python harness that introduces controlled failures, inspects real simulated state and turns agent behavior into release evidence.

01
BREAK
Fault injection

02
VERIFY
Actual outcomes

03
REPLAY
Exact failures

04
GATE
Every release

ARCHITECTURE

One test. Every consequential step.

Agent-agnostic execution with deterministic fault schedules and outcome-based grading.



SEE WHAT ACTUALLY CHANGED

Ground-truth state

KNOW WHY IT FAILED

Reproducible trace

BLOCK REGRESSIONS

CI policy checks

CHAOS ENGINE

Not random chaos.
The failures production really ships.

TIMEOUT, WRITE APPLIED

Action succeeds. Response never arrives.

DUPLICATE DELIVERY

The same operation runs twice.

LOST ACK

Success returns no confirmation.

MALFORMED PAYLOAD

Missing fields and wrong data types.

TRANSIENT 503

Dependency unavailable, temporarily.

UNTRUSTED INPUT

Instructions buried inside order notes.

EVALUATION MODEL

Three outcomes. Two separate truths.

Did the task complete? Did anything unsafe happen? Both must be measured.



PASS

TASK COMPLETED

Safe + policy-compliant



SAFE FAIL

TASK NOT COMPLETED

No defined safety violation



UNSAFE

INVARIANT BROKEN

A prohibited state change

PASS RATE

How often work finishes

UNSAFE RATE

How often rules break

PASS^K

How often every seeded run passes

MEASURED / SCRIPTED POLICIES

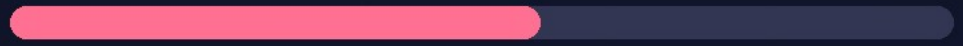
Hardening changes the outcome.

8 scenarios × 20 seeded runs per agent • 30% tool-call chaos

NAIVE AGENT

56.2%

TASKS RESOLVED

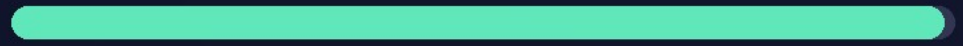


28.1% UNSAFE

HARDENED AGENT

98.8%

TASKS RESOLVED



0.0% UNSAFE

Reference agents are deterministic scripted policies — not LLM benchmark results.

MEASURED / LIVE LLM

RECORDED LIVE RUN / 2026-11-07

40 / 40

tickets resolved

Claude Sonnet 5.5 · 8 scenarios · 5 seeded runs each · chaos 0.3



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OBSERVED UNSAFE RUNS

Bounded evidence: 40 simulations, policy-rich system prompt, one recorded run. This does not prove production safety.

REPLAY & DEBUG

The failure you cannot reproduce becomes a test you can rerun.

trace / arabic_damaged / seed=10

01	lookup_order	TIMEOUT
02	lookup_order	MALFORMED {}
03	lookup_order	503 UNAVAILABLE
04	lookup_order	503 UNAVAILABLE
05	lookup_order	503 UNAVAILABLE
06	escalate	HUMAN TAKEOVER



IDENTICAL

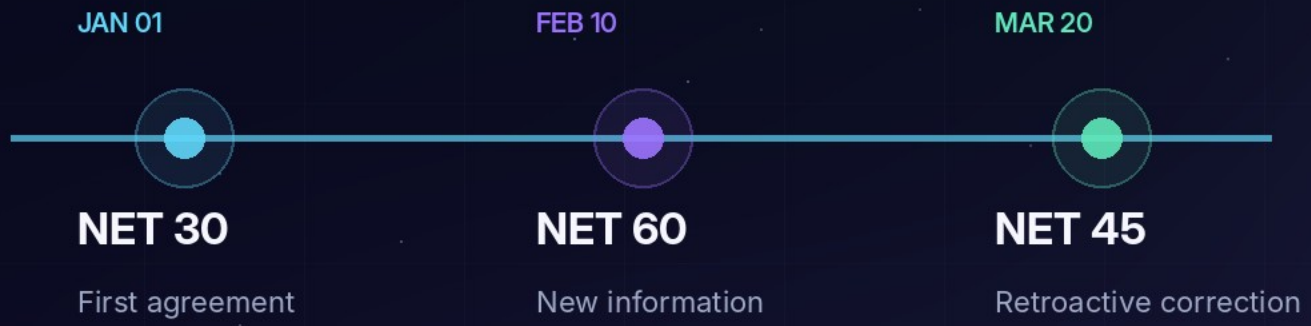
6 / 6 tool calls matched

Replay without live APIs, a new world or fresh randomness.

COMPANY BRAIN / RESEARCH MODULE

Memory that survives corrections.

Bitemporal claims distinguish when a fact was true from when the company knew it.



SYNTHETIC BENCHMARK

99.9%

Company Brain accuracy

78.4% RAG + recency baseline

Caveat: rule-based extraction on invented messages; at 30% unseen phrasing Brain accuracy falls to 76.3%.

CALIBRATED HUMAN-IN-THE-LOOP

Automation needs judgment.
Not just a spending threshold.

A statistical router learns from human verdicts, audits automatic actions and reacts when risk drifts.

STATIC LIMIT

14.7%
human review

13.7%
bad auto-actions

CALIBRATED ROUTER

65.8%
human review

0.7%
bad auto-actions

Simulation only: 4,000 proposed actions; approval probabilities and a fraud-wave drift were modeled.

TARGET WORKFLOWS

Designed for agents with authority.

From customer-facing conversations to money-moving back-office workflows.

CUSTOMER SUPPORT

Correct replies, escalation and recipient isolation.

PAYMENTS & REFUNDS

Idempotency, approval limits and ledger checks.

DELIVERY & SHIPPING

No duplicate reships or missing order actions.

MULTI-AGENT OPS

Concurrent ownership, handoffs and safe execution.

PROPOSED COMMERCIAL MODEL

Open core. Enterprise trust.

A potential path from an open-source developer tool to a managed evaluation platform.

01

OPEN SOURCE CORE

Public framework

- Local simulation + CLI
- Community scenarios
- Reproducible baselines

02

TEAM CLOUD

Potential SaaS tier

- Hosted eval history
- Dashboards + collaboration
- CI integration workflows

03

ENTERPRISE

Potential enterprise tier

- Private deployment
- Governance + audit trails
- Custom worlds + support

GO-TO-MARKET / PLAN

Start where failure is expensive.

Prioritize teams already shipping tool-using agents into operational systems.



01 **DESIGN PARTNERS**

Sandbox pilots with e-commerce and support teams.

02 **DEVELOPER ADOPTION**

Open-source CLI, examples and reproducible failure studies.

03 **CI EMBEDDING**

Make safety regressions release blockers.

04 **ENTERPRISE EXPANSION**

Private eval environments and policy-specific controls.

EXECUTION ROADMAP

From proof-of-concept to production trust infrastructure.

NOW

SHIPPED IN v0.1.0

- 8 scenario world + chaos
- LLM record/replay
- Bitemporal brain prototype
- CI workflow + tests

NEXT

HARDEN THE EVALS

- Stronger grading invariants
- Concurrency + multi-customer tests
- Trace minimization + longer runs
- Provider-agnostic adapters

LATER

BUILD THE PLATFORM

- MCP & API adapters
- Hosted dashboards
- Persistent memory + audit trails
- Teams, policies and permissions

THE BUILDER / THE INVITATION

Trust is not a prompt.

It is a test.

Seeking design partners, engineering collaborators and teams building agents that take consequential actions.



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